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Inventor(s)

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### **Systems and methods for providing singulation of objects for processing using object movement redistribution**

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#### **Abstract**

A processing system including a singulation system is disclosed. The singulation system includes a conveying system for moving objects to be processed from a source area along a first direction, a detection system for detecting objects at the conveying system, and for selecting certain selected objects for redistribution on the conveying system, and a movement redistribution system for redistributing the certain selected objects on the conveying system for providing a singulated stream of objects.

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## **Background/Summary**

**PRIORITY (1)** The present application is a continuation of U.S. patent application Ser. No. 16/998,049, filed Aug. 20, 2020; which is a continuation of U.S. patent application Ser. No. 15/961,294, filed Apr. 24, 2018, now U.S. Pat. No. 10,792,706, issued Oct. 6, 2020, which claims priority to U.S. Provisional Patent Application Ser. No. 62/489,121, filed Apr. 24, 2017, the disclosures of which are hereby incorporated by reference in their entireties.

## **BACKGROUND**

(1) The invention generally relates to automated, robotic and other processing systems, and relates in particular to automated and robotic systems intended for use in environments requiring, for example, that a variety of objects (e.g., articles, parcels or packages) be processed, e.g., sorted and/or otherwise distributed to several output destinations.

(2) Many object distribution systems receive objects in a disorganized stream that may be provided as individual objects or objects aggregated in groups such as in bags, arriving on any of several different conveyances, commonly a conveyor, a truck, a pallet, a Gaylord, or a bin. Each object must then be distributed to the correct destination container, as determined by identification information associated with the object, which is commonly determined by a label printed on the object. The destination container may take many forms, such as a bag or a bin.

(3) The processing of such objects has traditionally been done, at least in part, by human workers that scan the objects, e.g., with a hand-held barcode scanner, and then place the objects at assigned locations. For example many order fulfillment operations achieve high efficiency by employing a process called wave picking. In wave picking, orders are picked from warehouse shelves and placed at locations (e.g., into bins) containing multiple orders that are sorted downstream. At the processing stage individual objects are identified, and multi-object orders are consolidated, for example into a single bin or shelf location, so that they may be packed and then shipped to

customers. The processing (e.g., sorting) of these objects has traditionally been done by hand. A human sorter picks an object from an incoming bin, finds a barcode on the object, scans the barcode with a handheld barcode scanner, determines from the scanned barcode the appropriate bin or shelf location for the article, and then places the article in the so-determined bin or shelf location where all objects for that order have been defined to belong. Automated systems for order fulfillment have also been proposed. See for example, U.S. Patent Application Publication No. 2014/0244026, which discloses the use of a robotic arm together with an arcuate structure that is movable to within reach of the robotic arm.

(4) Other ways of identifying objects by code scanning either require manual processing, or require that the code location be controlled or constrained so that a fixed or robot-held code scanner (e.g., barcode scanner) can reliably detect it. Manually operated barcode scanners are generally either fixed or handheld systems. With fixed systems, such as those used at point-of-sale systems, the operator holds the object and places it in front of the scanner so that the barcode faces the scanning device's sensors, and the scanner, which scans continuously, decodes any barcodes that it can detect. If the object is not immediately detected, the person holding the object typically needs to vary the position or rotation of the object in front of the fixed scanner, so as to make the barcode more visible to the scanner. For handheld systems, the person operating the scanner looks for the barcode on the object, and then holds the scanner so that the object's barcode is visible to the scanner, and then presses a button on the handheld scanner to initiate a scan of the barcode.

(5) Further, many current distribution center sorting systems generally assume an inflexible sequence of operations whereby a disorganized stream of input objects is first singulated into a single stream of isolated objects presented one at a time to a scanner that identifies the object. A conveyance element or elements (e.g., a conveyor, a tilt tray, or manually movable bins) transport the objects to the desired destination or further processing station, which may be a bin, a chute, a bag or a conveyor, etc.

(6) In conventional parcel sortation systems, human workers or automated systems typically retrieve objects in an arrival order, and sort each object into a collection bin based on a set of given heuristics. For instance, all objects of like type might go to a collection bin, or all objects in a single customer order, or all objects destined for the same shipping destination, etc. The human workers or automated systems are required to receive objects and to move each to their assigned collection bin. If the number of different types of input (received) objects is large, a large number of collection bins is required.

(7) Such a system has inherent inefficiencies as well as inflexibilities since the desired goal is to match incoming objects to assigned collection bins. Such systems may require a large number of collection bins (and therefore a large amount of physical space, large capital costs, and large operating costs) in part, because sorting all objects to all destinations at once is not always most efficient.

(8) Current state-of-the-art sortation systems rely on human labor to some extent. Most solutions rely on a worker that is performing sortation, by scanning an object from an induction area (chute, table, etc.) and placing the object in a staging location, conveyor, or collection bin. When a bin is full, another worker empties the bin into a bag, box, or other container, and sends that container on to the next processing step. Such a system has limits on throughput (i.e., how fast can human workers sort to or empty bins in this fashion) and on number of diverts (i.e., for a given bin size, only so many bins may be arranged to be within efficient reach of human workers).

(9) Other partially automated sortation systems involve the use of recirculating conveyors and tilt trays, where the tilt trays receive objects by human sortation (human induction), and each tilt tray moves past a scanner. Each object is then scanned and moved to a pre-defined location assigned to the object. The tray then tilts to drop the object into the location. Further, partially automated systems, such as the bomb-bay style recirculating conveyor, involve having trays open doors on the bottom of each tray at the time that the tray is positioned over a predefined chute, and the object is

then dropped from the tray into the chute. Again, the objects are scanned while in the tray, which assumes that any identifying code is visible to the scanner.

(10) Such partially automated systems are lacking in key areas. As noted, these conveyors have discrete trays that can be loaded with an object; they then pass through scan tunnels that scan the object and associate it with the tray in which it is riding. When the tray passes the correct bin, a trigger mechanism causes the tray to dump the object into the bin. A drawback with such systems however, is that every divert requires an actuator, which increases the mechanical complexity and the cost per divert can be very high.

(11) An alternative is to use human labor to increase the number of diverts, or collection bins, available in the system. This decreases system installation costs, but increases the operating costs. Multiple cells may then work in parallel, effectively multiplying throughput linearly while keeping the number of expensive automated diverts at a minimum. Such diverts do not ID an object and cannot divert it to a particular spot, but rather they work with beam breaks or other sensors to seek to ensure that indiscriminate bunches of objects get appropriately diverted. The lower cost of such diverts coupled with the low number of diverts keep the overall system divert cost low.

(12) Unfortunately, these systems don't address the limitations to total number of system bins. The system is simply diverting an equal share of the total objects to each parallel manual cell. Thus each parallel sortation cell must have all the same collection bins designations; otherwise an object might be delivered to a cell that does not have a bin to which that object is mapped. There remains a need for a more efficient and more cost effective object sortation system that sorts objects of a variety of sizes and weights into appropriate collection bins or trays of fixed sizes, yet is efficient in handling objects of such varying sizes and weights.

#### SUMMARY

(13) In accordance with an embodiment, the invention provides a processing system including a singulation system. The singulation system includes a conveying system for moving objects to be processed from a source area along a first direction, a detection system for detecting objects at the conveying system, and for selecting certain selected objects for redistribution on the conveying system, and a movement redistribution system for redistributing the certain selected objects on the conveying system for providing a singulated stream of objects.

(14) In accordance with another embodiment, the invention provides a singulation system including a conveying system for moving objects to be sorted from a source area along a first direction, a selection system for selecting certain selected objects for redistribution on the conveying system, and a movement redistribution system for redistributing the certain selected objects on the conveying system and by returning the certain selected objects to an earlier stage of the conveying system such that a singulated stream of objects may be provided to an object processing system.

(15) In accordance with a further embodiment, the invention provides a method of providing singulation of objects. The method includes the steps of moving objects to be sorted from a source area along a first direction of a conveying system, detecting objects at the conveying system, selecting certain selected objects for redistribution on the conveying system, and redistributing the certain selected objects on the conveying system to provide a singulated stream of objects.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following description may be further understood with reference to the accompanying drawings in which:

(2) FIG. 1 shows an illustrative diagrammatic view of an object singulation processing system in accordance with an embodiment of the present invention;

(3) FIG. 2 shows an illustrative diagrammatic view of the object singulation processing system of

FIG. 1 at later point in time;

(4) FIG. 3 shows an illustrative diagrammatic view of the object singulation processing system of FIG. 1 at a further later point in time;

(5) FIG. 4 shows an illustrative diagrammatic view of the processing conveyor of FIG. 1;

(6) FIG. 5 shows an illustrative diagrammatic view of an object processing system in accordance with another embodiment of the present invention;

(7) FIG. 6 shows an illustrative diagrammatic front view of the drop perception system of FIG. 1;

(8) FIG. 7 shows an illustrative diagrammatic rear view of the drop perception system of FIG. 1;

(9) FIG. 8 shows an illustrative diagrammatic view of an object distribution system in accordance with an embodiment of the invention;

(10) FIG. 9 shows an illustrative diagrammatic view of a shuttle wing sortation system of FIG. 8;

(11) FIG. 10 shows an illustrative top view of a destination location in the shuttle wing sortation system of FIG. 9;

(12) FIG. 11 shows an illustrative diagrammatic view of an object distribution system in accordance with another embodiment of the invention;

(13) FIG. 12 shows an illustrative diagrammatic view of an object distribution system in accordance with a further embodiment of the invention;

(14) FIG. 13 shows an illustrative diagrammatic view of a portion of the shuttle wing processing system of FIG. 12;

(15) FIG. 14 shows an illustrative diagrammatic view of the portion of the shuttle wing processing system of FIG. 13, with an object being dropped from the carriage;

(16) FIG. 15 shows an illustrative diagrammatic view of an object distribution system in accordance with yet a further embodiment of the invention;

(17) FIG. 16 shows an illustrative diagrammatic view of a portion of the shuttle wing processing system of FIG. 15;

(18) FIG. 17 shows an illustrative diagrammatic view of the portion of the shuttle wing processing system of FIG. 16, with an object being dropped from the carriage;

(19) FIG. 18 shows an illustrative diagrammatic view of the object distribution system of FIG. 15 showing a destination bin position sensor;

(20) FIG. 19 shows an illustrative diagrammatic view of a flowchart showing selected processing steps in a system in accordance with an embodiment of the present invention; and

(21) FIG. 20 shows an illustrative diagrammatic view of a flowchart showing bin assignment and management steps in a system in accordance with an embodiment of the present invention.

(22) The drawings are shown for illustrative purposes only.

#### DETAILED DESCRIPTION

(23) In accordance with an embodiment, the invention provides a processing system that includes an input system for receiving a wide variety of objects to be processed, and a singulation system for providing a singulated stream of objects for efficient processing of the objects. In further embodiments, the system may include an identification system for identifying objects, and an output system for providing the singulated stream of objects at desired output destinations.

Individual parcels need to be identified and conveyed to desired parcel-specific locations. The described systems reliably automate the identification and conveyance of such parcels, employing in certain embodiments, a set of conveyors and sensors and a robot arm. In short, applicants have discovered that when automating sortation of objects, there are a few main things to consider: 1) the overall system throughput (parcels sorted per hour), 2) the number of diverts (i.e., number of discrete locations to which an object can be routed), 3) the total area of the sortation system (square feet), and 4) the annual costs to run the system (man-hours, electrical costs, cost of disposable components).

(24) Processing objects in a distribution center (e.g., sorting or order fulfillment) are applications for automatically identifying and moving objects. In a shipping distribution center for example,

objects commonly arrive in trucks, are conveyed to sortation stations where they are processed (e.g., sorted) according to desired destinations, aggregated in bags, and then loaded in trucks for transport to the desired destinations. Another application would be in the shipping department of a retail store or order fulfillment center, which may require that objects be processed for transport to different shippers, or to different distribution centers of a particular shipper. In a shipping or distribution center the objects may take form of plastic bags, boxes, tubes, envelopes, or any other suitable container, and in some cases may also include objects not in a container. In a shipping or distribution center the desired destination is commonly obtained by reading identifying information printed on the object or on an attached label. In this scenario the destination corresponding to identifying information is commonly obtained by querying the customer's information system. In other scenarios the destination may be written directly on the object, or may be known through other means.

(25) In accordance with various embodiments, therefore, the invention provides a method of taking individual objects from a disorganized stream of objects, providing a singulated stream of objects, identifying individual objects, and processing them to desired destinations. The invention further provides methods for loading objects into the system, for conveying objects from one point to the next, for excluding inappropriate or unidentifiable objects, for grasping objects, for determining grasp locations, for determining robot motion trajectories, for transferring objects from one conveyor to another, for aggregating parcels and transferring to output conveyors, for digital communication within the system and with outside information systems, for communication with human operators and maintenance staff, and for maintaining a safe environment.

(26) Important components of an automated object identification and processing system, in accordance with an embodiment of the present invention, as shown in FIG. 1. FIG. 1 shows a system **10** that includes an infeed hopper **12** into which objects **14** may be dumped, e.g., by a dumper or transferred from a Gaylord. An infeed conveyor **16** conveys objects from the infeed hopper **12** to a primary conveyor **19**. The infeed conveyor **16** may include cleats **18** for assisting in lifting the objects **14** from the hopper **12** onto the primary conveyor **19**. Primary perception system **32** surveys the objects **14** to identify objects when possible, to determine good grasp points, and to select certain objects for repositioning on the conveyor **19** in accordance with various embodiments of the invention.

(27) The system also includes one or more programmable motion systems **20**, **24** such as robotic arms **21**, **25**, each of which includes a gripper **22**, **26** for gripping objects **14**. Each robotic arm may be equipped with sensors and computing, that when combined is assumed herein to exhibit the following capabilities: (a) it is able to pick objects up from a stream of (e.g., non-singulated) objects; (b) it is able to move the object to arbitrary places within its workspace; and, (c) it is able to generate a map of objects that it is able to pick, represented as a candidate set of grasp points in the workcell, and as a list of polytopes enclosing the object in space. The allowable objects are determined by the capabilities of the robotic system. Their size, weight and geometry are assumed to be such that the robotic system is able to pick, move and place them.

(28) With further reference to FIGS. 2 and 3, the robotic arms **21**, **25** are used to move objects either to the beginning of the conveyor **19**, or to positions that will provide a singulated stream of objects **15**. For example, FIGS. 1-3 show that an object **13** may be picked up by the robotic arm **21** and moved to an upstream portion of the conveyor **19**, and show that an object **17** may be picked up by the other robotic arm **25** and also moved to an upstream portion of the conveyor **19**. The movement of the objects **13**, **17** provides a singulated stream of objects **15** by removing selected objects and returning the objects to an upstream portion of the conveyor.

(29) Significantly, a singulated stream of objects **15** is provided (as shown at **30**), and this singulated stream of objects may be monitored by a perception system **33**, and may be delivered to an identification system **34** (such as a drop perception system as discussed below) as a singulated stream and without requiring that a robotic system place objects into the drop perception system.

Objects may then fall through the system **34** onto, for example, a conveyor system **36**, for further processing as discussed below. By providing a singulated stream of objects for processing, the system is able to more effectively control the object processing rate, and reduce the incidence of errors that may occur, for example if two objects in close contact with each other are perceived as being one object. The infeed conveyor **16** may also be in communication with the controller **78** and the robotic arm **24**, and the speed of the infeed conveyor **16** may be adjusted to either slow down if moving too fast, or speed up if system determines that more bandwidth exists for a faster input. The speed and direction of the conveyor **19** may also be adjusted as may be necessary to provide the singulate stream of objects.

(30) As further shown in FIG. **4**, the system may monitor the movement of the conveyor **19**, enabling the system to maintain dynamic information regarding divided areas or zones of a defined distance (Z.sub.d) such that the singulated stream of objects is provided with one object **15** per area Z.sub.d. This may be achieved, for example, by moving objects **9**, **7**, **5** and **4** to upstream positions on the conveyor and may include moving objects **8**, **6** to locations that are more central to a respective area Z.sub.d. In certain embodiments and applications, the system may move objects to downstream positions on the conveyor in the process of providing a singulated stream of objects.

(31) As further shown in FIG. **5**, a system **10'** of another embodiment of the invention may include a primary conveyor **29** with cleats **31**. The cleats may, for example, define the divided areas or zones such that when the singulated stream of objects **15** is provided, it is provided with one object positioned within each cleated area. The primary conveyor **29** is otherwise similar to and functions like primary conveyor **19** of FIGS. **1-4**, and the cleats **31** may be similar to the cleats **18** of the infeed conveyor **16**. The remaining elements of the system of FIG. **5** are the same as those of FIGS. **1**, **2** and **3** and bear the same reference numerals.

(32) The correct processing destination is determined from the symbol (e.g., barcode) on the object. It is assumed that the objects are marked in one or more places on their exterior with a visually distinctive mark such as a barcode or radio-frequency identification (RFID) tag so that they may be identified with a scanner. The type of marking depends on the type of scanning system used, but may include 1D or 2D barcode symbologies. Multiple symbologies or labeling approaches may be employed. The types of scanners employed are assumed to be compatible with the marking approach. The marking, either by barcode, RFID tag, or other means, encodes a symbol string, which is typically a string of letters and numbers, that identifies the object.

(33) The perception system **34** may be supported by stands or may be suspended from above. As further shown in FIGS. **6** and **7**, the perception system **34** may include a structure **52** having a top opening **54** and a bottom opening **56**, and may be covered by an enclosing material **58**. The structure **52** includes a plurality of sources (e.g., illumination sources such as LEDs) **60** as well as a plurality of image perception units (e.g., cameras) **62**. The sources **60** may be provided in a variety of arrangements, and each may be directed toward the center of the opening. The perception units **62** are also generally directed toward the opening, although some cameras are directed horizontally, while others are directed upward, and some are directed downward. The system **34** also includes an entry source (e.g., infrared source) **64** as well as an entry detector (e.g., infrared detector) **66** for detecting when an object has entered the perception system **34**. The LEDs and cameras therefore encircle the inside of the structure **52**, and the cameras are positioned to view the interior via windows that may include a glass or plastic covering (e.g., **68**).

(34) An aspect of certain embodiments of the present invention, is the ability to identify via barcode or other visual markings of objects by employing a perception system into which objects may be dropped. Automated scanning systems would be unable to see barcodes on objects that are presented in a way that their barcodes are not exposed or visible to a single camera. The system **34** therefore is designed to view an object from a large number of different views very quickly, reducing or eliminating the possibility of the system **34** not being able to view identifying indicia on an object.

(35) Key features in the perception system are the specific design of the perception system so as to maximize the probability of a successful scan, while simultaneously minimizing the average scan time. The probability of a successful scan and the average scan time make up key performance characteristics. These key performance characteristics are determined by the configuration and properties of the perception system, as well as the object set and how they are marked.

(36) The two key performance characteristics may be optimized for a given item set and method of labeling. Parameters of the optimization for a system include how many scanners, where and in what orientation to place them, and what sensor resolutions and fields of view for the scanners to use. Optimization can be done through trial and error, or by simulation with models of the object.

(37) Optimization through simulation employs a scanner performance model. A scanner performance model is the range of positions, orientations and barcode element size that an identifying symbol can be detected and decoded by the scanner, where the barcode element size is the size of the smallest feature on the symbol. These are typically rated at a minimum and maximum range, a maximum skew angle, a maximum pitch angle, and a minimum and maximum tilt angle.

(38) Typical performance for camera-based scanners are that they are able to detect symbols within some range of distances as long as both pitch and skew of the plane of the symbol are within the range of plus or minus 45 degrees, while the tilt of the symbol can be arbitrary (between 0 and 360 degrees). The scanner performance model predicts whether a given symbol in a given position and orientation will be detected.

(39) The scanner performance model is coupled with a model of where symbols would expect to be positioned and oriented. A symbol pose model is the range of all positions and orientations, in other words poses, in which a symbol will expect to be found. For the scanner, the symbol pose model is itself a combination of an article gripping model, which predicts how objects will be held by the robotic system, as well as a symbol-item appearance model, which describes the possible placements of the symbol on the object. For the scanner, the symbol pose model is itself a combination of the symbol-item appearance model, as well as an inbound-object pose model, which models the distribution of poses over which inbound articles are presented to the scanner. These models may be constructed empirically, modeled using an analytical model, or approximate models may be employed using simple sphere models for objects and a uniform distribution over the sphere as a symbol-item appearance model.

(40) The operations of the systems described herein are coordinated by the central control system **78** as shown in FIGS. **1-3, 5, 8** and **10**. The central control system is comprised of one or more workstations or central processing units (CPUs). The correspondence between barcodes, for example, and outbound destinations is maintained by the central control system in a database called a manifest. The central control system maintains the manifest by communicating with a warehouse management system (WMS).

(41) If the perception system successfully recognizes a marking on the object, then the object is then identified and forwarded to a sorting station or other processing station. If the object is not identified, the robotic system may divert the object to a human sortation bin **76** to be reviewed by a human.

(42) With reference to FIG. **8**, in a processing system **100** of an embodiment of the invention, objects **14** passing through the secondary perception unit **34** fall onto secondary conveyor **36**. Diverters **70** including push bars **72** divert objects to shuttle sections **74** as appropriate. While only two such diverters and shuttle sections are shown, any number of such diverters and shuttle sections may be used. Unidentified objects or otherwise unacceptable objects continue along secondary conveyor **36** and fall into secondary exception bin **76**. The diverters **70** are in communication with the controller **78**, which is in communication with the scanner **32** as well as the indexing position of the conveyor **19**. Once an object falls through the scanner and lands on the conveyor **36**, the system notes the conveyor position of the object. The scanner information is

processed, and the object (if identified) is associated with that conveyor location, and its processing location is identified (as discussed in more detail below). As the conveyor advances, the system will know when the object is in the line of activation of a selected diverter, and will activate the diverter to push the object into the appropriate carriage. The carriage then moves the object to the assigned bin as discussed in more detail below. In various embodiments, the diverters may push an object off through various other ways, such as using a robot or a diverting guide, and in further embodiments, the diverters may pull an object off of the conveyor.

(43) As further shown with reference to FIG. 9, each shuttle section **74** includes a carriage **80** that shuttles back and forth between destination chutes **84** that include guide walls **85** that lead to two rows of bins **90, 92** on either side of track **88**. Again, a central computing and control station **78** communicates with other computers distributed in the other components, and also communicates with the customer information system, provides a user interface, and coordinates all processes. As shown in FIG. 9, each processing bin **90, 92** of each shuttle section **74** may include a pull out drawer **82** from which each of the two opposing processing bins (e.g., **90, 92**) may be accessed and emptied. Each pull-out drawer **82** may also include light indicators **94** to indicate when the processing bin (e.g., **90, 92**) is either full or is ready to be emptied based on system heuristics, e.g., that the bin is statistically unlikely to receive another object soon. In other embodiments, such lights may be positioned above the respective bin. Each drawer may also include a lock **99** that a person must unlock to pull out the drawer **82**. The lock includes sensors that communicate with the controller **78**, and when a drawer is unlocked, the system knows not to sort to either bin in the unlocked drawer. This way, the system may continue operating while drawers are pulled and bins are emptied.

(44) As shown in FIG. 10, each bin (**90, 92**), may include one or more pairs of emitters **96** and sensors **98** at the top of the bin. Output from a sensor **98** that is representative of a prolonged interruption from the associated source, may be used to determine that the bin is full.

(45) FIG. 11 shows a processing system **100'** similar to that shown in FIG. 8 (where the identical components have the same reference numbers), except that the shuttle sections **74'** of FIG. 8 are positioned alongside (parallel with) the conveyor **36'**. In particular, a first diverter **70'** may push an object into a carriage **80'** at one end of a shuttle section **74'**, while a second diverter **70''** may push an object into a carriage **80''** in the middle of a shuttle section **74''**. In accordance with further embodiments, many different arrangements are possible, and each is within the spirit and scope of the present invention. Each drawer **82'** and **82''** may be discussed herein and the indicator lights **84', 84''** may be located above the drawers **82', 82''**.

(46) Similarly, the diverters **70', 70''** are in communication with the controller **78**, which is in communication with the scanner **34** as well as the indexing position of the conveyor **36'**. Again, in various embodiments, the diverters may push an object off through various other ways, such as using a robot or a diverting guide, and in further embodiments, the diverters may pull an object off of the conveyor. Once an object falls through the scanner and lands on the conveyor, the system notes the conveyor position of the object. The scanner information is processed, and the object (if identified) is associated with that conveyor location, and its processing location is identified (as discussed in more detail below). Again, as the conveyor advances, the system will know when the object is in the line of activation of a selected diverter, and will activate the diverter to push the object into the appropriate carriage. The carriage then moves the object to the assigned bin as discussed in more detail below.

(47) FIG. 12 shows a processing system **200** similar to that of systems **100, 100'** (with similar elements bearing similar reference numerals), except that the system **200** includes carriages **101** that ride along a track (e.g., a circular track) **102**. When a carriage **101** is positioned below the drop scanner **34**, an object falls through the scanner and is identified as discussed above. The carriage **101** is then moved between rows of bins **104**. With further reference to FIGS. 13 and 14, when the carriage **101** is moved to a desired processing location, the carriage stops (or slows), and tilts to



dump the object **14** into the bin **104** (as shown in FIG. **14**) similar to the action of carriage **80** discussed above. Again, the object **14** may include indicia **15** such as a barcode that was detected by the scanner **34**. Similar to the embodiment of FIG. **9**, guide walls may be used to guide the object as it falls so that the object does not accidentally drop into a neighboring bin, and sensors (e.g., emitter/detector pairs) **96, 98** may be employed to detect when a bin is full as discussed above.

(48) FIG. **15** shows a processing system **200'** similar to that of systems **100, 100'** and **200** (with similar elements bearing similar reference numerals), except that the system **200'** includes carriages **202** that ride along a track (e.g., a circular track) **204**. When a carriage **202** is positioned below the drop scanner **34**, an object falls through the scanner and is identified as discussed above. The carriage **202** is then moved between rows of bins **206**, each of which may include, for example a pre-placed bag. With further reference to FIGS. **16** and **17**, when the carriage **202** is moved to a desired processing location, the carriage stops (or slows), and dumps the object **14** into the bin **206** (as shown in FIG. **16**) similar to the action of carriage **80** discussed above. Again, the object **14** may include indicia **15** such as a barcode that was detected by the scanner **34**.

(49) As further shown in FIG. **18**, when a bin **206** is full (e.g., by sensors as discussed above, or by the system knowing how many items are in the bin, or by having a human simply decide that a bin is full) a human may then pick up the bin **206**. Upon removing the bin **206**, a sensor system **208** under the bin will indicate that the bin (that specific bin) has been removed. The system may continue processing other bins, but will know not to sort to the removed bin. A new empty bin **210** may then be replaced on the opened location. Because the assignment of bin processing locations is dynamic and flexible, no further registration is required. As soon as the bin **210** is placed on the sensor **208**, the system will know that there is a new unassigned bin ready for dynamic processing as discussed further below.

(50) The assignment of processing bins may also be dynamic. For example, systems in accordance with further embodiments, provide improved transport and conveyor systems to provide a singulated stream of objects, and to provide dynamically changing patterns of object handling, with resulting efficiencies in the sortation process, lower space requirements, lower demand for manual operations, and as a consequence, lower capital and operating costs for the entire system.

(51) During use, the sorting station may select an object and then identify the selected object by the perception system **32** (or may detect an identity of an object using a scanner on the articulated arm, or may use the robotic arm to move the object to a detection device). If the object has an assigned bin or a new bin is available, then the end effector will drop the object from the carriage into the bin. If the object is not identified the object may be dropped into a designated exception bin that is part of the shuttle wing, or the object may continue traveling in the carriage **202** along the track **204** and later be dropped into an exception bin **76** (e.g., as discussed above with reference to FIG. **8**).

(52) The system assigns a bin to the object if a new bin is available and the object is not yet assigned a bin at that sorting station. What is significant is that the sorting station is not pre-assigned a large set of collection bins assigned to all possible objects that may appear in the input path. Further, the central controller may employ a wide variety of heuristics that may further shape the process of dynamically assigning objects to collection bins as discussed in more detail below. Once bins are either filled or otherwise completed, the completed bins are signaled as being done and ready for further processing (e.g., by lights **92** associated with bin **90, 92** in FIG. **10**).

(53) As shown in FIG. **19**, a sortation process of the invention at a sorting station may begin (step **400**) by providing a singulated stream of objects that, one at a time, drop an object into the drop scanner (step **402**). The system then identifies the new object (step **404**). The system then will determine whether the object is yet assigned to any collection bin (step **406**). If not, the system will determine whether a next bin is available (step **408**). If no next bin is available (step **410**), the robotic system will return the object to the input buffer (step **410**) and return to step **402**.

Alternatively, the system can pick one of the collection bins that is in process and decide that it can

be emptied to be reused for the object in hand, at which point the control system can empty the collection bin or signal a human worker to do it. If a next bin is available (and the system may permit any number of bins per station), the system will then assign the object to a next bin (step **412**). The system then places the object into the assigned bin (step **414**). The system then returns to step **402** until finished. Again, in certain embodiments, the secondary conveyor may be an indexed conveyor that moves in increments each time an object is dropped onto the conveyor. The system may then register the identity of the object, access a warehouse manifest, and determine an assigned bin location or assign a new bin location.

(54) A process of the overall control system is shown, for example, in FIG. **20**. The overall control system may begin (step **500**) by permitting a new collection bin at each station to be assigned to a group of objects based on overall system parameters (step **502**) as discussed in more detail below. The system then identifies assigned bins correlated with objects at each station (step **504**), and updates the number of objects at each bin at each station (step **506**). The system then determines that when a bin is either full or the system expects that the associated sorting station is unlikely to see another object associated with the bin, the associated sorting station robotic system will then place the completed bin onto an output conveyor, or signal a human worker to come and empty the bin (step **508**), and then return to step **502**.

(55) Systems of various embodiments provide numerous advantages because of the inherent dynamic flexibility. The flexible correspondence between sorter outputs and destinations provides that there may be fewer sorter outputs than destinations, so the entire system may require less space. The flexible correspondence between sorter outputs and destinations also provides that the system may choose the most efficient order in which to handle objects, in a way that varies with the particular mix of objects and downstream demand. The system is also easily scalable, by adding sorters, and more robust since the failure of a single sorter might be handled dynamically without even stopping the system. It should be possible for sorters to exercise discretion in the order of objects, favoring objects that need to be handled quickly, or favoring objects for which the given sorter may have a specialized gripper.

(56) In various embodiments therefore, the object processing system may include a carriage **80** that shuttles back and forth on a track between destination bins. A central computing and control station **78** communicates with other computers distributed in the other components, and also communicates with the customer information system, provides a user interface, and coordinates all processes. In other embodiments, the system may include a track (e.g., in a loop) along which carriages may travel in one direction past a plurality of destination bins.

(57) Systems of various embodiments provide numerous advantages because of the inherent dynamic flexibility. The flexible correspondence between sorter outputs and destinations provides that there may be fewer sorter outputs than destinations, so the entire system may require less space. The flexible correspondence between sorter outputs and destinations also provides that the system may choose the most efficient order in which to handle objects, in a way that varies with the particular mix of objects and downstream demand. The system is also easily scalable, by adding sorters, and more robust since the failure of a single sorter might be handled dynamically without even stopping the system. It should be possible for sorters to exercise discretion in the order of objects, favoring objects that need to be handled quickly, or favoring objects for which the given sorter may have a specialized gripper.

(58) The system provides in a specific embodiment an input system that interfaces to the customer's conveyors and containers, stores objects for feeding into the system, and feeds those objects into the system at a moderate and controllable rate. In one embodiment, the interface to the customer's process takes the form of a dumper from a Gaylord, but many other embodiments are possible. In one embodiment, feeding into the system is by an inclined cleated conveyor with overhead flow restrictors, e.g., baffles. In accordance with certain embodiments, the system feeds objects in at a modest controlled rate. Many options are available, including variations in the conveyor slope and

speed, the presence, size and structure of cleats and baffles, and the use of sensors to monitor and control the feed rate.

(59) The system includes in a specific embodiment a primary perception system that monitors the stream of objects on the primary conveyor. Where possible the primary perception system may identify the object to speed or simplify subsequent operations. For example, knowledge of the objects on the primary conveyor may enable the system to make better choices regarding which objects to move to provide a singulated stream of objects.

(60) Those skilled in the art will appreciate that numerous modifications and variations may be made to the above disclosed embodiments without departing from the spirit and scope of the present invention.

## Claims

1. A method of processing objects, said method comprising: indiscriminately engaging and lifting a subset of a plurality of in-feed objects from an in-feed contained area by a first conveyor; indiscriminately depositing the subset of the plurality of in-feed objects by the first conveyor at a first end of a second conveyor; detecting by a first perception unit the subset of the plurality of in-feed objects on the second conveyor as the second conveyor moves in a first direction and providing in-feed perception data responsive thereto; controlling the speed of movement of the first conveyor by a controller responsive to the in-feed perception data; selecting certain selected objects of the plurality of in-feed objects detected by the first perception unit for redistribution on the second conveyor; redistributing the certain selected objects on the second conveyor by moving the certain selected objects, using an end effector of a programmable motion device, to provide a singulated stream of objects on the second conveyor, each object of the singulated stream of objects being provided one per a predefined zone on the second conveyor, said predefined zone being provided as one of a plurality of zones; passing each of the objects of the singulated stream of objects from the second conveyor through a drop scanner towards a third conveyor, the drop scanner including a plurality of perception units that provide singulated perception data for identifying each of the objects of the singulated stream of objects; and directing each of the objects of the singulated stream of objects by a diverter toward a destination location of a plurality destination locations based on the singulated perception data.

2. The method of claim 1, wherein the in-feed contained area includes a hopper.

3. The method of claim 1, wherein the first conveyor includes a cleated conveyor.

4. The method of claim 1, wherein the second conveyor moves the certain selected objects in discrete increments.

5. The method of claim 1, wherein redistributing the certain selected objects includes moving an object upstream on the second conveyor from a current location into one of the plurality of zones.

6. The method of claim 1, wherein redistributing the certain selected objects includes moving an object toward a center of one of the plurality of zones on the second conveyor.

7. The method of claim 1, wherein the plurality of destination locations comprises opposing rows of processing bins and wherein each processing bin is provided in a pull-out drawer.

8. The method of claim 1, wherein passing each of the objects through the drop scanner includes dropping the objects of the singulated stream of objects to fall through the drop scanner.

9. The method of claim 8, wherein the drop scanner includes a housing defining an opening therethrough, and wherein the method further includes providing the singulated perception data from the plurality of perception units of the drop scanner for identifying each object of the singulated stream of objects dropped through the drop scanner onto the third conveyor.

10. The method of claim 9, wherein the plurality of perception units includes a plurality of cameras, a plurality of scanners, or a combination thereof.

11. A method of processing objects, said method comprising: indiscriminately engaging and lifting

a subset of a plurality of in-feed objects from an in-feed contained area by a first conveyor; indiscriminately depositing the subset of the plurality of in-feed objects by the first conveyor at a first end of a second conveyor; detecting by a first perception unit the subset of the plurality of in-feed objects on the second conveyor as the second conveyor moves in a first direction and providing in-feed perception data responsive thereto; selecting certain selected objects of the plurality of in-feed objects detected by the first perception unit for redistribution on the second conveyor; redistributing the certain selected objects on the second conveyor by moving the certain selected objects, using an end effector of a programmable motion device, to provide a singulated stream of objects on the second conveyor, each object of the singulated stream of objects being provided one per a predefined zone on the second conveyor, said predefined zone being provided as one of a plurality of zones; passing each object of the singulated stream of objects from the second conveyor through a drop scanner towards a third conveyor, the drop scanner including a plurality of perception units that provide singulated perception data for identifying each object of the singulated stream of objects; and directing each object of the singulated stream of objects by a diverter toward a destination location based on the singulated perception data.

12. The method of claim 11, wherein the in-feed contained area includes a hopper.

13. The method of claim 11, wherein the first conveyor includes a cleated conveyor.

14. The method of claim 11, wherein the second conveyor moves the certain selected objects in discrete increments.

15. The method of claim 11, wherein redistributing the certain selected objects includes moving an object upstream on the second conveyor from a current location into one of the plurality of zones.

16. The method of claim 11, wherein redistributing the certain selected objects includes moving an object toward a center of one of the plurality of zones on the second conveyor.

17. The method of claim 11, wherein the plurality of destination locations comprises opposing rows of processing bins and wherein each processing bin is provided in a pull-out drawer.

18. The method of claim 11, wherein passing each of the objects through the drop scanner includes dropping the objects of the singulated stream of objects to fall through the drop scanner.

19. The method of claim 18, wherein the drop scanner includes a housing defining an opening therethrough, and wherein the method further includes providing the singulated perception data from the plurality of perception units of the drop scanner for identifying each object of the singulated stream of objects dropped through the drop scanner onto the third conveyor.

20. The method of claim 19, wherein the plurality of perception units includes a plurality of cameras, a plurality of scanners, or a combination thereof.

21. An object processing system comprising: an in-feed contained area providing a plurality of in-feed objects; a first conveyor for indiscriminately engaging and lifting a subset of the plurality of in-feed objects from the in-feed contained area; a second conveyor including a first end onto which the subset of the plurality of in-feed objects are indiscriminately deposited; a first perception unit for detecting the subset of the plurality of in-feed objects on the second conveyor as the second conveyor moves in a first direction and for providing in-feed perception data responsive thereto; a controller for selecting certain selected objects of the plurality of in-feed objects for redistribution on the second conveyor; a programmable motion device for redistributing the certain selected objects on the second conveyor by moving the certain selected objects, using an end effector of the programmable motion device, to provide a singulated stream of objects on the second conveyor, each of the objects of the singulated stream of objects being provided one per a predefined zone on the second conveyor, said predefined zone being provided as one of a plurality of zones; a drop scanner for receiving each object of the singulated stream of objects as the singulated stream of objects are provided towards a third conveyor, the drop scanner including a plurality of perception units that provide singulated perception data for identifying each object of the singulated stream of objects; and a diverter for directing each of the objects of the singulated stream of objects toward a destination location of a plurality destination locations based on the singulated perception data.

22. The object processing system of claim 21, wherein the in-feed contained area includes a hopper.
  23. The object processing system of claim 21, wherein the first conveyor includes a cleated conveyor.
  24. The object processing system of claim 21, wherein the second conveyor moves the certain selected objects in discrete increments.
  25. The object processing system of claim 21, wherein the plurality of destination locations comprises opposing rows of processing bins and wherein each processing bin is provided in a pull-out drawer.
  26. The object processing system of claim 21, wherein the programmable motion device moves an object toward a center of one of the plurality of zones on the second conveyor.
  27. The object processing system of claim 26, wherein the drop scanner includes a housing defining an opening therethrough, and wherein the drop scanner provides perception data from the plurality of perception units of the drop scanner for identifying each object of the singulated stream of objects dropped through the drop scanner onto the third conveyor.
  28. The object processing system of claim 21, wherein the programmable motion device moves an object upstream on the second conveyor from a current location into one of the plurality of zones.
  29. The object processing system of claim 28, wherein each object of the singulated stream of objects fall through the drop scanner.
  30. The object processing system of claim 29, wherein the plurality of perception units includes a plurality of cameras, a plurality of scanners, or a combination thereof.
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